

Source panel	Ministry of the Environment General Waste Treatment Survey FY2005-FY2024 23,599 facility-year rows
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analytical split

A. Adoption frame: entry margin	
Sample	Coded at-risk facilities first observed without power generation 13,770 facility-years 2,035 facilities 141 events
Estimand	First observed reporting of power generation in the following fiscal year
Model	Lagged discrete-time logit hazard with year and prefecture fixed effects
Role	Identifies selective observed entry, not a causal retrofit mechanism

B. Generator frame: performance margin	
Sample	Operating generators with positive throughput and power output 5,683 observations 1,016 facilities
Outcome	Electricity recovered per tonne clipped and logged MWh/t
Model	Pooled, year-FE, and random-effects panel specifications
Role	Describes structured performance hierarchy within generators

Synthesis	The same national fleet is interpreted through two linked but non-identical margins; one average-fleet estimate would conflate entry with post-entry performance.
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